

JOSÉ ANTONIO MORA AYALA

Computer Systems Engineer | Frontend Engineer | Team Lead

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PROFESSIONAL SUMMARY

Computer Systems Engineer and Frontend Engineer with experience in architecture and technical leadership of enterprise web applications. Specialized in React and TypeScript, with expertise designing scalable architectures, leading teams, migrating platforms from Vue 2 to React, and implementing testing practices and code quality standards. Experienced in integrating AI into products and development processes, as well as collaborating on defining business logic and UX/UI design.

PROFESSIONAL EXPERIENCE

Pro Meritum

Team Lead Frontend | Jan. 2026 – Present

- Lead a multidisciplinary team of 5 members, including Frontend developers and UX/UI designers, coordinating task assignments, code reviews, and technical standards definition.
- Led the architectural design for V2, migrating the platform from Vue 2 to React and establishing Feature-Sliced Design (FSD) to organize features, components, and reusable logic under a modular and scalable structure.
- Structured data consumption flows through a centralized Axios and TanStack Query configuration, implementing a Query Factory pattern to standardize queries, mutations, caching, and data invalidation.
- Established a standardized development environment through JSDoc documentation, dependency audits with pnpm audit, and automated testing using Husky, ESLint, Vitest, and React Testing Library.
- Designed and implemented an RBAC system manageable from the UI, integrating route authorization on the Frontend and permission validation on the Backend.
- Designed and integrated AI-assisted development workflows using Claude and agentic tools, creating skills and optimizing prompts and contexts to reduce the team's technical research time.

Frontend Engineer | Oct. 2024 – Dec. 2025

- Gradually assumed technical coordination of the team starting October 2025, managing deliverables and unifying development criteria.
- Maintained and optimized version 1 of the platform in Vue 2, resolving critical bugs and refactoring tightly coupled legacy components.
- Collaborated with UX/UI and stakeholders to define user flows, prototypes, and business rules, translating complex requirements into functional specifications.
- Implemented a dark/light theme system using Tailwind CSS and a custom component library.

FEATURED PROJECTS

Recallio - Collaborative Learning & Spaced Repetition Platform

- Built a full-stack solution integrating Astro, React, and NestJS through a monorepo structure based on pnpm workspaces, optimizing dependency management and deployments.
- Designed the database schema with PostgreSQL and Prisma for integrating the SM-2 algorithm,

simultaneously managing public repositories and private datasets.

- Established a community collaboration flow managed via an administrative dashboard with RBAC for evaluating, approving, and reviewing contributions.

Personal Finance & Shared Expenses Platform

- Conceptualized and developed a shared expense management application to track credit card statement closing dates, payment deadlines, and group budgets.
- Implemented business logic and entities using Next.js and NestJS.
- Developed an OCR pipeline to extract and normalize information from PDF bank statements, validate transactions, and automatically assign individual debts, including recurring charges and interest-free monthly installment purchases.

ESCOMIA - Capstone Project

- Led the Frontend architecture and implementation of a PWA for managing university procedures using React, TanStack Query, Vite, Tailwind CSS, and Shadcn UI.
- Developed RESTful APIs with Express.js and integrated OpenAI to implement automated comment moderation and a conversational chatbot tailored to academic requirements and deadlines.

EDUCATION

National Polytechnic Institute (IPN) - School of Computing (ESCOM)

B.S. in Computer Systems Engineering | Aug. 2020 – Aug. 2024

- Distinction: Academic Excellence / Graduated with Honors.
- GPA: 9.6 / 10.

Center for Scientific and Technological Studies No. 2 "Miguel Bernard" - IPN

Technician in Digital Graphic Design | 2017 – 2020

- GPA: 9.8 / 10.

TECHNICAL SKILLS

- **Core:** TypeScript, JavaScript, React, HTML, CSS.
- **Frameworks:** Next.js, Vue 2/3, Astro.
- **Architecture & State Management:** Feature-Sliced Design, TanStack Query, TanStack Table, Zustand.
- **UI:** Tailwind CSS, Shadcn UI, Bootstrap.
- **Backend:** Node.js, NestJS, Express.js, Hapi, REST APIs, MongoDB, PostgreSQL, MySQL.
- **Testing & Quality:** Vitest, React Testing Library, MSW, ESLint, Husky, Commitlint.
- **Tools & DevOps:** Git, GitHub, GitLab, Docker, Vite, Axios, pnpm.
- **AI:** Claude, OpenAI, Agentic workflows.
- **Design:** Figma, Illustrator, Photoshop.

Languages: Spanish (Native), English (B2).